

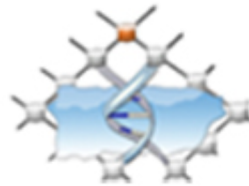


University of Puerto Rico



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**Puerto Rico - IDeA Networks of Biomedical Research
Excellence**



PRINBRE
IDeA Network of Biomedical Research Excellence

Python3 Part 1 – Crash Course in Python3 for Future STEM Coders

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- The following material is the result of PR-INBRE research and curriculum development effort to provide a set of educational materials for research training and curriculum changes in biology programs across the island to support PR-INBRE efforts to establish a Community of Practice in Bioinformatics that offers a fruitful environment to increase computational and bioinformatics skills among traditional researchers and students (undergraduate and graduate) in the island. They have been developed as a part of the NIH funded project “**Puerto Rico IDeA Network Biomedical Research Excellence (PRINBRE)**” (Award Number 5P20GM103475).
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- Most recent versions of these presentations can be found at <http://inbre.hpcf.upr.edu/>.





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Competencies

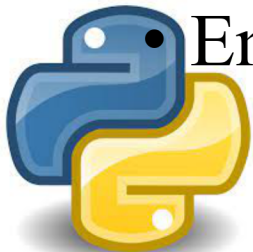
Record and write simple and common **Python scripts to deal with Bioinformatic needs for biological data analyses** using a Jupyter Notebook





Objectives

- Use **Google Colab as an environment** to practice and learn common **Python** lines of code.
- Formulate simple calculations using **Python**
- Identity the following **datatypes**: integer, float, and string



- Employ **variables** for different datatypes



Objectives

- Manipulate integer, float, and string datatypes
- Use **bioinformatic tools** to find a protein-coding sequence
- **Evaluate a protein-coding sequence** of interest and calculate its GC content.



Debug code



Target Audience

- This training is addressed to beginners, highly motivated (eager to learn what is needed in order to be competitive without the tendency to self-limit when learning computational skills by saying that is difficult) wanting to become familiar with the **Python** programming language and become the Script Master of their bioinformatic analysis.





The importance of Python

- R is great; however, you'll eventually run into the Python language
- No matter your field, python will be a necessity (chemistry, biology, engineering, ecology, mathematics, etc)
- Knowing Python gives you an edge when job searching and navigating graduate research
- The majority of bioinformatic tools are python-based and will lead you navigate under the hood code to successfully execute said program's purpose.





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Lesson 1.0.0

Connecting to **Google** **Colab** to learn common **Python** lines of code



(Gmail account ready for colab.research.google.com)

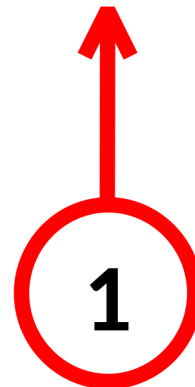


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Google Colab

<https://colab.research.google.com>



Activate the link

Activate colab.research.google.com

The screenshot displays the Google Colaboratory (Colab) web interface in a browser. The address bar shows colab.research.google.com. The page has a header with the Colab logo, "Welcome To Colaboratory", and a menu (File, Edit, View, Insert, Runtime, Tools, Help). On the right of the header are links for "Share", settings, and a user profile. A left sidebar contains a "Table of contents" with links to "Getting started", "Data science", "Machine learning", "More Resources", "Featured examples", and a "Section" button. The main content area is titled "Welcome to Colab!" and includes a paragraph: "If you're already familiar with Colab, check out this video to learn about interactive tables, the executed code history view, and the command palette." Below this is a video player with a play button and a thumbnail showing a man and the text "3 Cool Google Colab Features". Further down, the section "What is Colab?" explains that Colab allows writing and executing Python in the browser, listing benefits: "Zero configuration required", "Access to GPUs free of charge", and "Easy sharing". It also mentions that Colab is suitable for students, data scientists, and AI researchers, and provides a link to "Introduction to Colab". The "Getting started" section begins with the text: "The document you are reading is not a static web page, but an interactive environment called a **Colab notebook** that lets you write and execute code." The page number "10" is visible in the bottom right corner of the content area.

colab.research.google.com

Welcome To Colaboratory

File Edit View Insert Runtime Tools Help

Share

Table of contents

- Getting started
- Data science
- Machine learning
- More Resources
- Featured examples
- Section

Welcome to Colab!

If you're already familiar with Colab, check out this video to learn about interactive tables, the executed code history view, and the command palette.

3 Cool Google Colab Features

What is Colab?

Colab, or "Colaboratory", allows you to write and execute Python in your browser, with

- Zero configuration required
- Access to GPUs free of charge
- Easy sharing

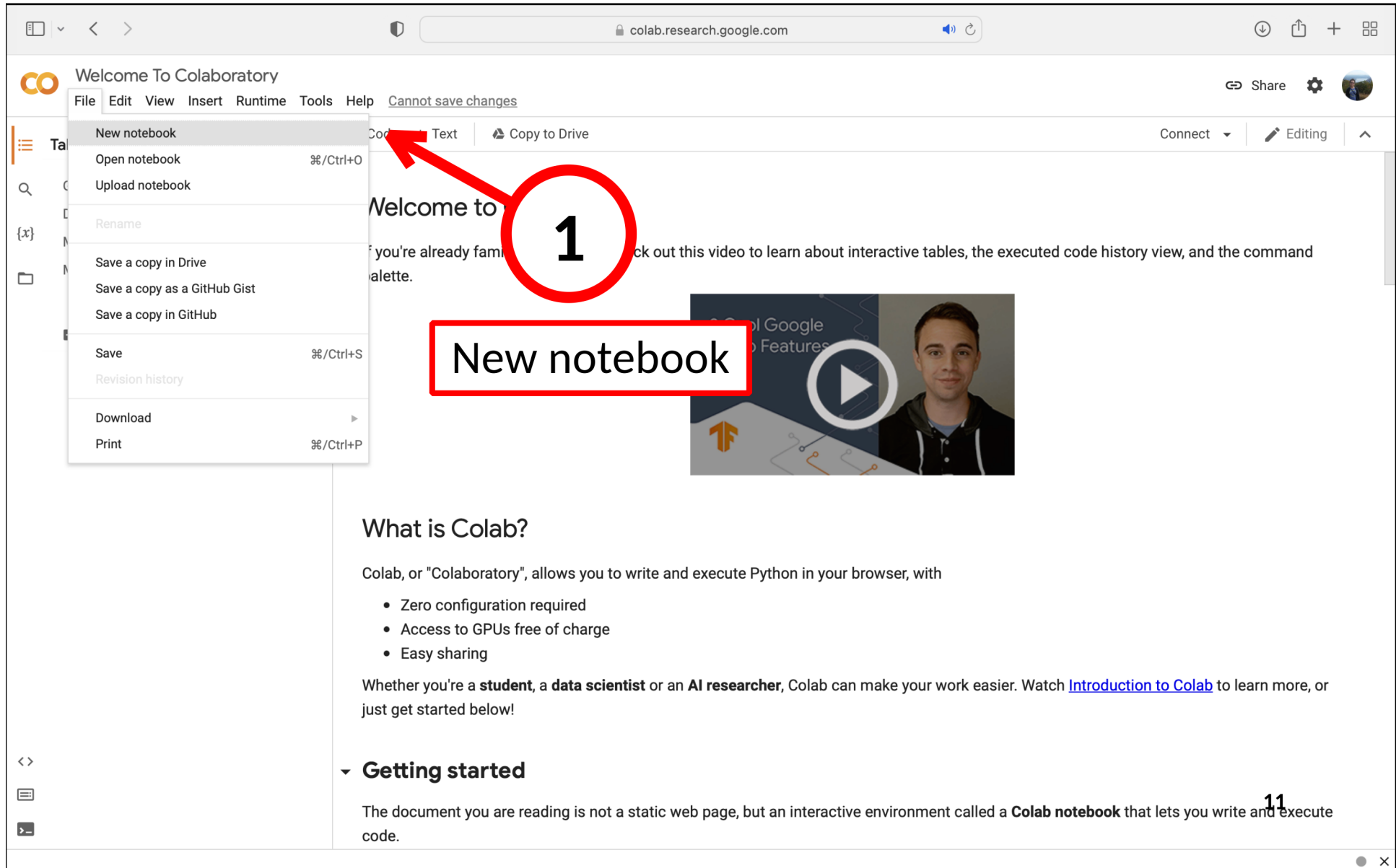
Whether you're a **student**, a **data scientist** or an **AI researcher**, Colab can make your work easier. Watch [Introduction to Colab](#) to learn more, or just get started below!

Getting started

The document you are reading is not a static web page, but an interactive environment called a **Colab notebook** that lets you write and execute code.

10

Open a New notebook



colab.research.google.com

Welcome To Colaboratory

File Edit View Insert Runtime Tools Help Cannot save changes

New notebook
Open notebook ⌘/Ctrl+O
Upload notebook
Rename
Save a copy in Drive
Save a copy as a GitHub Gist
Save a copy in GitHub
Save ⌘/Ctrl+S
Revision history
Download
Print ⌘/Ctrl+P

Code Text Copy to Drive

Connect Editing

1

New notebook

What is Colab?

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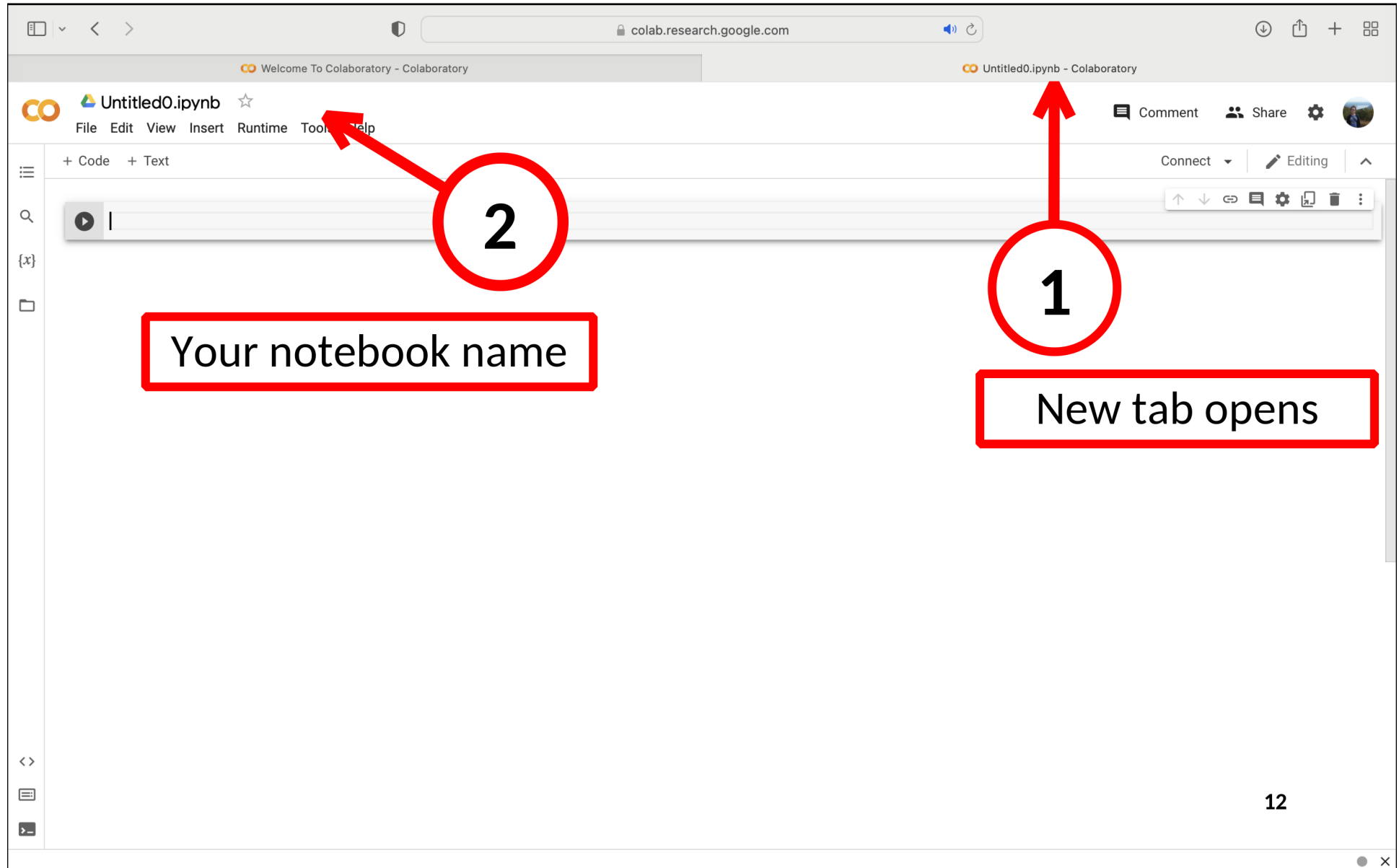
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▼ **Getting started**

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11

Welcome to your new notebook!





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https://github.com/bioinfwithjudith/PR-INBRE_Python_Workshop

3



In a new tab, activate the link

This is a GitHub repository!

The screenshot shows a web browser window displaying a GitHub repository. The browser's address bar shows 'github.com'. The repository name 'PR-INBRE_Python_Workshop' is visible, along with the user 'bioinwithjudith'. The repository is public and has 0 stars, 0 forks, and 1 watch. The main branch is 'main'. The repository contains several files and folders: 'PR-INBRE_2023', 'PR-INBRE_2024', 'README.md', and 'Student_PART_1_Python3_Workshop.ipynb'. The 'README.md' file is selected, showing the title 'PR-INBRE Python Workshop 2025' and a description of the workshop. The right sidebar contains sections for 'About', 'Releases', 'Packages', and 'Languages'. The 'Languages' section shows 'Jupyter Notebook' at 100.0%.

bioinwithjudith / PR-INBRE_Python_Workshop

Code Issues Pull requests Actions Projects Security Insights Settings

PR-INBRE_Python_Workshop Public

Unpin Unwatch 1 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file Add file Code

bioinwithjudith 2025 update e209c88 · 12 minutes ago 70 Commits

PR-INBRE_2023	2025 update	12 minutes ago
PR-INBRE_2024	2025 update	12 minutes ago
README.md	2025 update	13 minutes ago
Student_PART_1_Python3_Workshop.ipynb	updated files for students	last year

README

PR-INBRE Python Workshop 2025

This workshop is a crash course designed for beginner python coders in the STEM field. The development of sequencing methods and decreasing costs leads to the need of scientists knowing how to code. Not only is it important for scientists to know how to develop and execute code but scientists may run into looking at another scientists code and being able to decipher what's going on for their own research benefit.

Part 1 - The use of a Python Interpreter and running your first set of commands.

About

This workshop was developed for PR-INBRE.

- Readme
- Activity
- 0 stars
- 1 watching
- 0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Languages

- Jupyter Notebook 100.0%

Download Jupyter Notebook

The screenshot shows the GitHub repository page for 'PR-INBRE_Python_Workshop' by user 'bioinfwithjudith'. The repository is public and has 0 forks and 0 stars. The file list shows several files, including 'Student_PART_1_Python3_Workshop.ipynb', which is highlighted with a red arrow pointing to a red circle containing the number '4'. Below the file list, the README is visible, starting with 'PR-INBRE Python Workshop' and describing it as a crash course for beginner python coders. A red box with the word 'Click' is placed over the README text. The right sidebar contains sections for 'About', 'Releases', 'Packages', and 'Languages', with 'Languages' showing 'Jupyter Notebook' at 100.0%.

bioinfwithjudith / PR-INBRE_Python_Workshop

Code Issues Pull requests Actions Projects Security Insights Settings

PR-INBRE_Python_Workshop Public

Unpin Unwatch 1 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file Add file Code

bioinfwithjudith 2025 update e209c88 · 17 minutes ago 70 Commits

PR-INBRE_2023	2025 update	17 minutes ago
PR-INBRE_2024	2025 update	17 minutes ago
README.md	2025 update	19 minutes ago
Student_PART_1_Python3_Workshop.ipynb	updated files for students	last year

README

PR-INBRE Python Workshop

This workshop is a crash course designed for beginner python coders in the STEM field. The development of sequencing methods and decreasing costs leads to... how to code. Not only is it important for scientists to know how to develop and... s may run into looking at another scientists code and being able to decipher what's... ch benefit.

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Click

4

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Download Jupyter Notebook

The screenshot shows a GitHub repository for 'bioinfwithjudith / PR-INBRE_Python_Workshop'. The file 'Student_PART_1_Python3_Workshop.ipynb' is selected. The interface includes a sidebar with file navigation, a top navigation bar with repository tabs, and a main content area with tabs for 'Preview', 'Code', and 'Blame'. The 'Preview' tab is active, displaying the notebook content. A red arrow points to the 'Download raw file' button in the top right corner of the notebook preview area. This button is circled with a red '5'. Below the arrow, a red box contains the text 'Download notebook'.

bioinfwithjudith / PR-INBRE_Python_Workshop

PR-INBRE_Python_Workshop / Student_PART_1_Python3_Workshop.ipynb

bioinfwithjudith updated files for students 46960be · last year History

Download raw file

5

Download notebook

WELCOME TO THE FIRST DAY OF OUR PYTHON CRASH COURSE!

Lesson 1.1.0 Testing snippets of code using the Python Interpreter

Example 1

How many seconds(s) are in 10 milliseconds (ms)?

In []:

Problem 1

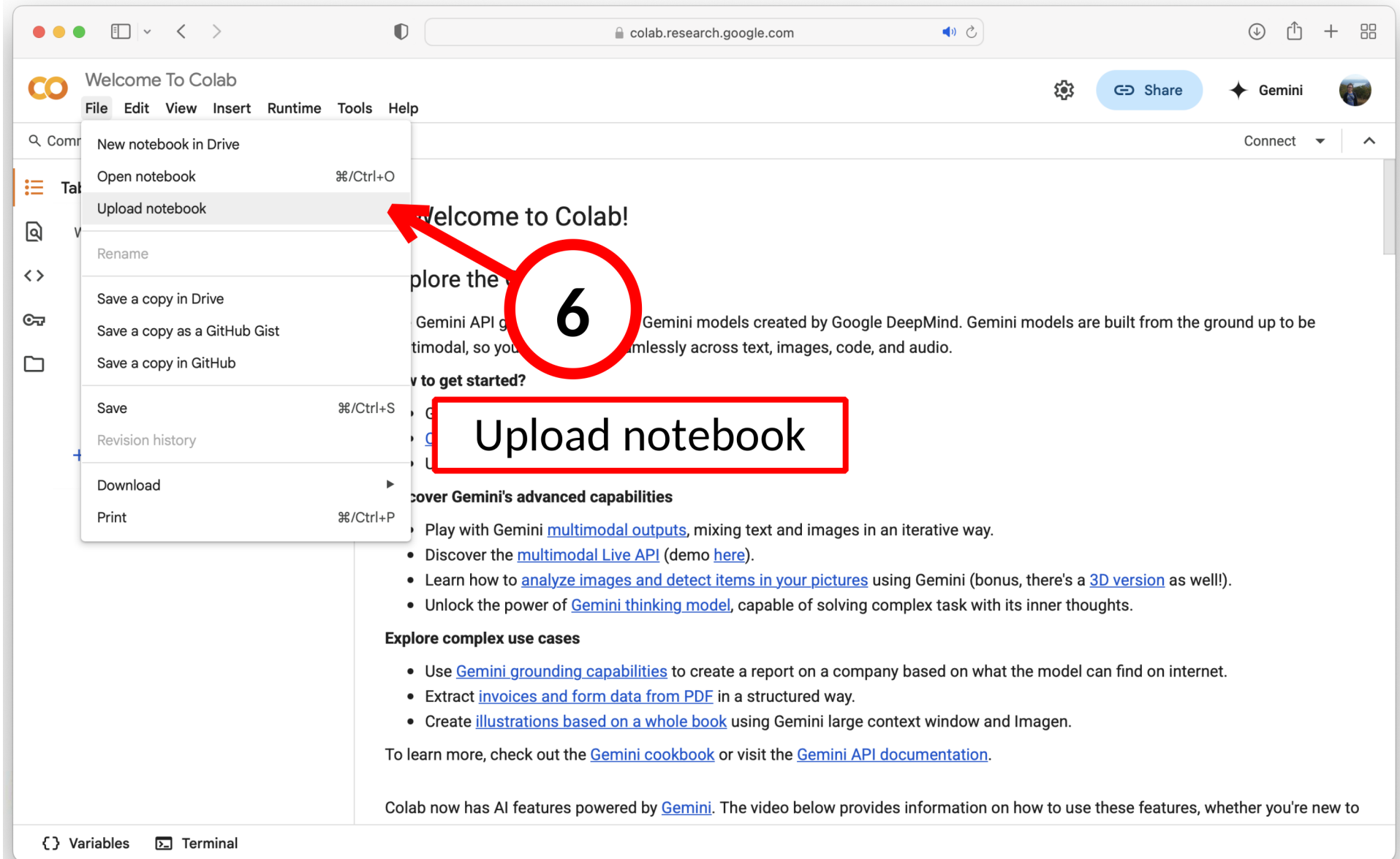
How many 5µL are in mL?

In []:

Problem 2

How many kg are in 0.0034g?

Upload existing **notebook**



Remember this for the next few days!

The screenshot shows a Google Colab notebook interface. The browser address bar displays `colab.research.google.com`. The notebook title is `Student_PART_1_Python3_Workshop.ipynb`. The top menu bar includes `File`, `Edit`, `View`, `Insert`, `Runtime`, `Tools`, and `Help`. On the right, there are icons for `Share`, `Gemini`, and a user profile. Below the menu bar, there are tabs for `Commands`, `+ Code`, and `+ Text`. A `Connect` button is also visible.

On the left side, a `Table of contents` panel is open, listing the following items:

- WELCOME TO THE FIRST DAY OF OUR PYTHON CRASH COURSE!**
- Lesson 1.1.0 Testing snippets of code using the Python Interpreter
 - Example 1
 - Problem 1
 - Problem 2
 - Problem 3
- Lesson 1.2.0. Python datatypes (Integers and floats and strings...oh my!)
 - Identify datatypes of Python objects using `type()` functions.
 - Run this classic string!
 - Assign a variable to this string and print variable
- Lesson 1.3.0 Evaluate the GC content of a DNA sequence
 - Create a variable for your favorite protein-coding gene sequence.
 - Get to know the CRTAM sequence

The main content area on the right displays the following lessons:

- WELCOME TO THE FIRST DAY OF OUR PYTHON CRASH COURSE!**
- › Lesson 1.1.0 Testing snippets of code using the Python Interpreter
 - [] ↳ 8 cells hidden
- › Lesson 1.2.0. Python datatypes (Integers and floats and strings...oh my!)
 - [] ↳ 8 cells hidden
- › Lesson 1.3.0 Evaluate the GC content of a DNA sequence
 - [] ↳ 10 cells hidden
- › Lesson 1.4.0 Running into errors
 - [] ↳ 1 cell hidden
- › Proficiency Assessment

At the bottom of the interface, there are tabs for `Variables` and `Terminal`.